Requirements Documentation

Interactive House - AI Smart Control and Safety Architecture

Revision History

Date	Version	Description	Author
2026-02-04	1.0	Initial automation-based functional requirements	Md Abu Taher
2026-02-24	1.1	Refined requirements to reflect AI-based smart control system (R1-R4)	Md Abu Taher
2026-02-24	1.1	Refined requirements to reflect AI-based smart control system (R5-R8)	Anitta Antony
2026-02-24	1.0	Smart Home Code	Daniel Marcarini
2026-02-24	1.1	Refined Requirements Based on R9	Daniel Holderik
2026-03-16	1.2	Updated AI requirements for web and Firebase AI	Md Abu Taher Anitta Antony
2026-03-16	1.1	Testing	Daniel Marcarini
2026-04-22	1.3	Added emergency button feature	Anitta Antony Md Abu Taher
2026-05-08	1.4	Added completion factors and clarified relations between requirements, design, and test plan for the final project meeting	Anitta Antony Md Abu Taher
2026-05-08	1.5	Added current automated Jest testing status for web authentication and Firebase configuration	Daniel Marcarini

Requirements List

Requirement Name	Priority	Progress	Completion factor
R1. Process natural language commands	Essential	Done	100%
R2. Control smart door and windows system	Essential	Done	100%
R3. Control ventilation system	Essential	Done	100%
R4. Monitor and respond to fire detection	Essential	Done	100%
R5. Monitor and respond to gas leakage	Essential	Done	100%
R6. Monitor environmental conditions	Essential	Done	100%
R7. Maintain device state information	Essential	Done	100%
R8. Provide intelligent feedback to user	Desirable	Done	100%

R9. Support Speech based user commands	Desirable		
R10. Support verification of web-based device control flow	Desirable	Done	100%
R11. Provide emergency assistance trigger	Essential	Done	100%

The completion factor shows the current implementation status of each requirement before the final project meeting. A requirement marked as 100% is implemented and ready for demonstration. A partially completed requirement means that the main functionality exists, but some limitations remain, such as dependency on Firebase configuration, authentication, or incomplete support across both web and mobile platforms

Contents

Requirements Descriptions

R1 – Process natural language commands

The system shall allow users to issue commands in natural language through the web and mobile application. The system shall interpret user intent, extract relevant information (action and target device), and convert it into structured executable instructions. The system shall handle variations in user phrasing and ambiguous expressions where possible.

R2 – Control smart door and windows system

The system shall support opening and closing the door and window system through AI commands. If lock functionality is available in the connected device state, the system may also support lock-related commands.

R3 – Control ventilation system

The system shall allow users to activate or deactivate the ventilation fan through AI commands. The System shall also allow the ventilation system to be triggered automatically in response to environmental or safety conditions.

R4 – Monitor and respond to fire detection

The system shall continuously monitor fire detection sensors. If fire is detected, the system shall automatically activate the fire alarm. The system shall notify the user of the detected emergency condition. Emergency responses shall occur without requiring user interaction.

R5 – Monitor and respond to gas leakage

The system shall continuously monitor gas sensors for leakage detection. If gas leakage is detected, the system shall automatically activate the gas alarm. The system may automatically activate the ventilation system to reduce risk. The system shall notify the user of the detected hazard.

R6 – Monitor environmental conditions

The system shall monitor indoor temperature and rain conditions. The system shall provide the user with the current temperature upon request. If environmental thresholds are exceeded, the system may suggest or trigger appropriate actions such as activating ventilation. If rain is detected, the system shall notify the user.

R7 – Maintain device state information

The system shall maintain updated state information for all connected devices and sensors. This includes door status, windows status, ventilation status, alarm states, rain detection status, and current temperature. The system shall use stored state information to correctly interpret context, dependent user commands.

R8 – Provide intelligent feedback to user

The system shall provide clear and meaningful responses to user commands. If a command is successfully executed, the system shall confirm the action. If a command is unclear or invalid, the system shall provide corrective feedback or request clarification.

R9 – Support speech-based user commands

The system shall allow users to issue commands using speech through the web and mobile application. Spoken input shall be converted into text using speech recognition and processed through the existing AI command processing system. The system shall support navigation and device control using voice commands.The system shall allow voice commands to trigger user interface actions such as pressing buttons, adjusting sliders, and navigating between application screens.

R10 – Support verification of web-based device control flow

The system shall support verification of web-based device-control interactions, including authentication, state updates, and synchronization between the web interface and backend device data. This requirement exists to ensure that implemented control features can be tested in a structured and repeatable way before demonstration and delivery.

The current implementation includes an initial automated web test suite using Jest and React Testing Library. The automated tests verify the authentication flow, signup flow, and Firebase configuration layer. Firebase Authentication and Firestore are mocked during the tests to make the verification repeatable and independent from the real backend state. The current automated test suite contains 14 passing tests across 4 test suites.

This partially supports the verification requirement by proving that important web application flows can be tested in a structured and repeatable way. Further automated tests are still needed for the hub page, device toggles, Firestore live synchronization, emergency flow, music page, and AI page.

R11 - Provide emergency assistance trigger

The system shall provide an emergency button accessible from the user interface across all application screens. When activated, the system shall trigger an emergency assistance flow, such as displaying a call interface or alert notification.

The emergency feature shall be designed for immediate access and visibility, ensuring usability for elderly or vulnerable users. The system shall simulate an emergency call or alert mechanism without requiring integration with external communication services.
Since the project does not integrate with real emergency services, the emergency assistance feature is implemented as a simulated emergency call or alert flow for demonstration purposes.

Relation with Test Plan

The requirements are related to the current web test plan mainly through authentication, navigation, hub state display, device-control actions, Firestore updates, and live synchronization. The test plan verifies login and signup flows, guest access, loading of existing Firestore data, device toggles for light, door, window, and fans, and automatic refresh of hub values through snapshot synchronization. Therefore, requirements such as R2, R3, R6, R7, R8, and R10 are connected to the documented test cases in the web test plan.

Some requirements are not directly covered by the written test cases in the current test plan. The AI command-processing requirement, speech-based command support, fire/gas emergency response, and the emergency assistance trigger were verified mainly through manual testing, implementation review, and demonstration during development. These requirements are still included in the final requirement artefact because they describe implemented or partially implemented project features, but future work should include separate written test cases for AI, speech input, safety responses, and the emergency button.

Relation with Design Document

The requirements are connected to the design document through the main components of the Interactive House system. The AI-related requirements, especially R1 and R8, are connected to D1, D2, D3, and D4, which describe the AI architecture, Firebase AI prompting rules, AI chat interface, error handling, reliability, and frontend state management. These design items explain how user commands are sent to Firebase AI, how responses are shown in the interface, and how the system handles loading states and errors.

The device-control requirements, especially R2, R3, and R7, are connected to the design through the hub interface, backend device-state handling, and Firestore synchronization. These parts of the design explain how device states such as door, window, fan, lights, and buzzer are read, updated, and kept consistent between the interface and backend data.

The speech-based command requirement, R9, is connected to D5, which describes speech recognition integration, voice navigation, floating voice control, and global state handling. The web verification requirement, R10, is connected to D6, which describes the web application testing strategy for authentication, navigation, Firestore-backed device toggles, and real-time synchronization. The emergency assistance requirement, R11, is connected to the emergency button interface and emergency call simulation design, where the user can access a simulated emergency flow from the application interface.

Daniel Marcarini - Smart Home code

Requirement Name	Priority
R1. Monitor and respond to gas detection	Essential
R2. Execute staged gas safety response	Essential
R3. Control smart window system	Essential
R4. Control smart door system	Essential
R5. Monitor environmental conditions	Essential
R6. Provide real-time system feedback	Essential
R7. Support manual user interaction	Essential
R8. Support Bluetooth remote interaction	Desirable

Requirements Descriptions

R1 – Monitor and respond to gas detection

The system shall continuously monitor gas sensor input values. If the measured gas level exceeds a predefined threshold, the system shall detect this condition as a potential hazard. The system shall initiate a safety response without requiring user interaction.

R2 – Execute staged gas safety response

When hazardous gas levels are detected, the system shall activate an audible alarm. The system shall execute a staged safety procedure including alarm activation and automatic ventilation support. The system shall maintain the alarm state until the hazardous condition is resolved.

R3 – Control smart window system

The system shall allow the smart window to be opened and closed using a servo motor. The system shall support both automatic window opening in emergency situations and manual control through user input.

R4 – Control smart door system

The system shall allow the smart door to be opened and closed using a servo motor. The system shall respond to user interaction commands and update the door state accordingly.

R5 – Monitor environmental conditions

The system shall continuously monitor indoor temperature and ambient light levels using appropriate sensors. The system shall provide environmental data for display and system logic processing.

R6 – Provide real-time system feedback

The system shall display system status information on an LCD screen. This shall include environmental conditions, alarm states, and device status information. The system shall provide clear feedback during both normal and emergency operation.

R7 – Support manual user interaction

The system shall allow users to interact with connected devices using physical push buttons. The system shall detect button state changes and execute corresponding device actions.

R8 – Support Bluetooth remote interaction

The system shall allow remote interaction through a Bluetooth communication interface. The system shall process incoming commands and execute corresponding actions when valid instructions are received.